Algorithm 1 Statics of the continuum legs

Initialize $\mathbf{q}^0$
for $i = 1, \ldots,$ maxiter **do**
  Form the residual: ▷ simulate states and gradients

$$\mathbf{b} \leftarrow \mathbf{F}(\mathbf{q}^{i-1}) + \mathbf{M}\mathbf{g} + \mathbf{F}_{ext} \tag{3}$$

  Compute $\mathbf{A}(\mathbf{q}_j^{i-1}) = \frac{\partial \mathbf{F}(\mathbf{q}^{i-1})}{\partial \mathbf{q}}$
  Solve the following system:

$$\mathbf{A}\, d\mathbf{q} = \mathbf{b} \tag{5}$$

  $\mathbf{q}^i \leftarrow \mathbf{q}^{i-1} + \alpha d\mathbf{q}$ ▷ solver step
  if $\|\mathbf{q}^i - \mathbf{q}^{i-1}\| < \epsilon$ **then**
    break ▷ solver converged
  end if
end for
return $\mathbf{q}^i$